

自动建模缺的不是代码

论文的判断是：Agent 已经会设计、编码和调参，但真正拉开方案质量的，是可用、可积累的 ML 工程经验。

AIBuildAI-2: A Knowledge-Enhanced Agent for Automatically Building AI Models

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AI models underpin data-centric applications such as image processing, text processing, clinical decision support, and increasingly the acceleration of discovery across the natural sciences such as biology, physics, and chemistry. Yet developing them remains a heavily manual process in which practitioners must design architectures, build training pipelines, and iteratively refine solutions, and it can be challenging for natural scientists without specialized AI engineering expertise to build the high-performing models their research demands. To reduce this burden and broaden access to AI for scientific discovery, agents that automatically build AI models have been proposed. However, the performance of these agents is largely limited by the parametric knowledge of their underlying large language models, which is static, often outdated, and sparse on practical AI model engineering know-how. To address this limitation, we introduce AIBuildAI-2, a knowledge-enhanced agent with an external, evolving knowledge system for automatically building AI models. The knowledge system of AIBuildAI-2 is hierarchical, organizing curated AI development knowledge into high-level knowledge instructions over logical categories and low-level knowledge documents under each category, from which the agent dynamically builds only the context relevant to its current

reliant on specialized expertise: practitioners must iteratively formulate modeling strategies, implement training pipelines, optimize hyperparameters, and refine their approach in light of empirical evidence (13–17). In particular, it can be challenging for natural scientists without specialized AI engineering expertise to build the high-performing models their research demands. To reduce this burden and broaden access to AI for scientific discovery, a growing line of work has developed autonomous AI agents that, given only a task description and training data, can produce competitive AI models end-to-end. Recent agents based on large language models (LLMs) (18–22) have demonstrated promising performance on realistic AI development benchmarks such as MLE-Bench (23), and increasingly approach the capability of experienced AI engineers on a broad range of Kaggle-style (24) tasks.

Despite this progress, the knowledge available to existing AI development agents during decision-making is fundamentally limited. The simplest agents rely solely on the parametric knowledge embedded in their underlying

论文原图 · 第 1 页

三种知识都不够

参数知识静态且可能过时；即时网页搜索噪声高、查询不稳定；固定文档库更可靠，却无法吸收新方法和历史运行经验。因此失败点不是单次不会回答，而是跨决策、跨任务没有稳定知识供应。

论文的核心命题

AIBuildAI-2 把外部知识接入设计、实现、调参与集成的每一步，并在任务结束后把执行轨迹重新写回知识库。作者希望把一次性搜索，改造成持续维护的工程记忆。

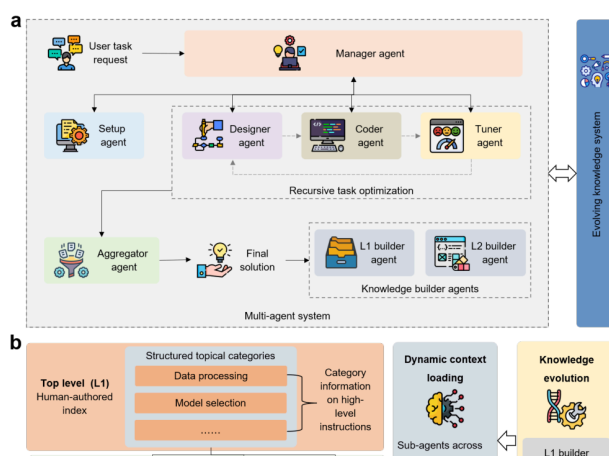


设计理解

编辑判断：这是一篇知识基础设施论文，而不是新模型架构论文。

多智能体如何搜索方案?

Manager 维护并行实验路线，再根据训练反馈决定下一步。



论文原图 · 第 3 页

角色不是简单流水线

Setup 准备环境，Designer 提方案，Coder 实现排错，Tuner 从日志调参，Aggregator 融合结果。Manager 可反复召回任一角色，因此实验失败会改变后续搜索方向。

七个候选仓库

每个任务初始化 7 个仓库，分别保存设计、代码、日志和验证分数。Manager 延长强路线、淘汰弱配置，把 24 小时预算集中到更有希望的方案。



设计理解

因为路线状态彼此隔离，系统能保留多样性，也能让最终集成利用不同误差模式。

两层知识库解决什么？

1000 份文档不是全部塞进上下文，而是先导航、后下钻：L1 决定去哪找，L2 回答具体怎么做。

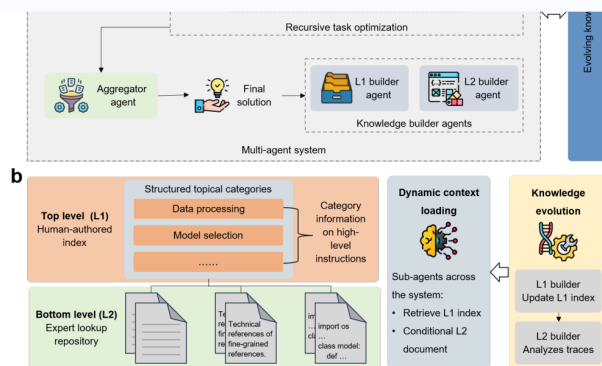


Fig. 1 | Overview of AIBuildAI-2. **a.** Overall workflow. AIBuildAI-2 takes as input a text description of an AI task and training data, and outputs trained model checkpoints and an inference script. It adopts a hierarchical sub-agent architecture: a setup agent prepares the software environment, a manager agent then oversees multiple parallel solution repositories and invokes a designer, coder, or tuner sub-agent on each, and an aggregator agent finally selects or ensembles the strongest repositories. All sub-agents are connected to a comprehensive, hierarchical, and continually evolving knowledge system, which they query for external expertise to ground their design, implementation, and tuning decisions. After each completed run, knowledge-builder agents distill the run into the knowledge system, so that it grows with the agent's experience. **b.** Knowledge system. The knowledge system has two levels: a top level (L1), a human-authored index of around 30 topical categories of AI model development (e.g., data processing, model selection, and ensemble techniques), each carrying a high-level knowledge instruction; and a bottom level (L2), an expert lookup repository of around 1,000 low-level knowledge documents that the L1 instructions point to for more specific knowledge. Through dynamic context loading, sub-agents across the system retrieve the L1 index and conditionally load L2 documents. The knowledge system is evolved by two builder agents: an L2 builder that analyzes execution traces to produce new L2 documents, and an L1 builder that incrementally updates the L1 instructions.

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L1：高层技能地图

人工定义约 30 个类别，覆盖视觉、NLP、时间序列、表格、分子数据、特征工程与集成。每个 L1[k] 同时包含该领域的高层指南，以及相关 L2 文档的短描述目录。

L2：场景化工程配方

约 1000 份自包含文档来自竞赛复盘、技术博客、官方文档、代码库和历史运行目录，保存某类特征提取、模型技巧或失败模式。分层结构降低无关知识进入上下文造成的干扰。



设计理解

信息流洞察：目录常驻，正文按需加载；知识规模和上下文长度因此可以解耦。

query(k,j) 到底做了什么？

论文没有复杂训练公式，最关键的“算法”是两阶段查询，它控制每次决策实际看到哪一段外部知识。

dex. The high-level knowledge instruction is a self-contained set of guidelines on how to approach category k , written in the spirit of agent skills (56), such as Anthropic Skills (57) and OpenAI Skills (58), so that it can be loaded directly into an LLM-based agent's context. The per-category L2 index lists, for each available document under k , the document identifier j paired with a short description summarizing what the document contains, allowing the agent to decide whether to retrieve a given document without first loading its full text. The actual low-level knowledge documents pointed to by these entries are stored in the L2 lookup, keyed by both category and document: $L_2[k][j]$ holds a single clean, self-contained document distilled from a group of related sources that all address one specific, narrowly scoped task or method within category k , such as competition write-ups, technical blog posts, open-source code, and library documentation, or distilled from the working directory of a previous AI-BuildAI-2 run.

The main agent and its sub-agents access this knowledge system through a single query tool that implements dynamic context loading. The L1 index is included in every agent invocation by default, so the agent always has a global view of available guidance. The tool exposes two call signatures. A category-level call $query(k) = L_1[k]$ loads the high-level knowledge instruction for category k together with its L2 index of pointers, giving the agent both the condensed guidance for k and the option to drill down further. After this category-level call returns, the agent can either proceed using the high-level knowledge instruction alone, when the condensed guidance in $L_1[k]$ is sufficient for the current decision, or, when it judges that

directory into a takeaway aimed at improving future runs on similar tasks; and (iii) produce a short description for retrieval. The result is the low-level knowledge document $L_2[k][j]$, stored under k with its description. These tasks demand strong reasoning over heterogeneous artifacts, motivating an LLM-based agent rather than a fixed pipeline.

The L1 builder takes (i) zero or one existing $L_1[k]$ for a target category k and (ii) one or more low-level knowledge documents under k , and emits an updated $L_1[k]$ comprising a refreshed high-level knowledge instruction and an updated per-category L2 index. The high-level knowledge instruction is prompted to be clean, self-contained, and to clearly instruct an agent on how to solve the AI tasks in category k or to carry out the specific AI method that k represents. When no L1 value exists yet for k , the L1 builder writes $L_1[k]$ from scratch using the low-level knowledge documents most relevant to k , identified by clustering document embeddings and verified via an LLM relevance check; when $L_1[k]$ already exists, the same agent edits the existing instruction in place and extends its L2 index. The human-expert-curated L1 index itself is held fixed across both the initial build and the subsequent evolution process.

Knowledge system initialization. To bootstrap L_2 , we crawl AI-development-related sources from Kaggle (24) competition forums, deep learning framework blogs (PyTorch (29), Hugging Face (60)), official documentation of widely used AI libraries (scikit-learn (61), Transformers (62)), open-source code repositories on GitHub (63), and selected arXiv (64) collections. We then apply a combination of MinHash-based deduplication (65) and an LLM-based rel-

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先问类别

执行 $query(k)=L1[k]$ ，返回类别 k 的高层指南和 L2 索引。若指南已足够，Agent 直接决策；这避免每个小问题都读取长文档，也保留了全局策略约束。

再问文档

需要窄场景配方时，Agent 根据索引描述选择 j ，再执行 $query(k,j)=L2[k][j]$ 。因此检索条件同时包含任务、当前 repository 状态和调用角色，而不是只用初始问题做一次向量搜索。



设计理解

它改变的是上下文中的信息流，不更新 LLM 参数；收益应来自更相关的证据进入正确决策时刻。

一次运行怎样变成记忆？

系统在任务结束后离线提炼，而不是把整段日志原样存入长期记忆。

Knowledge system evolution. The knowledge system continues to evolve after deployment through two complementary update streams. The first stream folds AIBuildAI-2's own experience back into the knowledge system: after each completed task, the L2 builder is run on the AIBuildAI-2 working directory. The second stream periodically incorporates newly published web content: as new competition write-ups, blog posts, library updates, and code repositories appear, the L2 builder is run on each new group of similar sources, so L_2 continues to track the state of the field rather than being frozen at the bootstrap snapshot. Each invocation in either stream produces an additional low-level knowledge document $L_2[k^*][j^*]$, stored under the most relevant category k^* and appended to $L_2[k^*]$ alongside the bootstrap corpus.

In both update streams, once a new low-level knowledge document $L_2[k^*][j^*]$ is added under category k^* , the L1 builder is re-invoked in evolution mode on k^* with the existing $L_1[k^*]$ and the new document as input, and emits an updated $L_1[k^*]$ in which the high-level knowledge instructions is refined to incorporate new strategies or amonate known pitfalls and the per-category L2 index is extended to point to the new document, while the original structure and length budget of $L_1[k^*]$ are preserved. This update procedure keeps the knowledge system well-formed and lightweight while ensuring that it accumulates increasingly comprehensive and battle-tested guidance derived from both the agent's own experience and the broader AI development community.

Blind Challenge data are publicly available at <https://huggingface.co/spaces/openadmet/OpenAdmet-ExpansionRx-Challenge>. The solution generated by AIBuildAI-2 for the heart disease prediction competition is available at <https://github.com/aibuildai/AI-Build-AI/tree/main/tasks/playground-series-s6e2>, and the solution generated by AIBuildAI-2 for the OpenAdmet ExpansionRx Blind Challenge is available at https://drive.google.com/drive/folders/1IC07fa3_QyDkhwgtpdtsjsE0FTaw18Tusg=sharing.

Code availability

The AIBuildAI-2 software is publicly available at <https://github.com/aibuildai/AI-Build-AI>, together with detailed instructions for using the system.

Author contributions

R.Z., P.Q., Q.C., and L.Z. contributed to conceptualization, methodology, software, investigation, analysis, writing—original draft, and writing—review and editing. F.X. contributed to conceptualization, methodology, investigation, analysis, writing—original draft, and writing—review and editing.

Competing interests

The authors declare no competing interests.

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L2 Builder 提炼经验

它读取完整工作目录，识别最相关类别，把设计选择、失败实验和最佳策略去重、归一化，生成新的 $L_2[k^*][j^*]$ 及检索描述。新增网页内容也经过同样的分类与清洗流程。

L1 Builder 压缩共识

新 L2 写入后，L1 Builder 在保持结构与长度预算的前提下，补充新策略或已知陷阱，并扩展索引。这样细节留在 L2，高频共识逐步上浮到 L1。



设计理解

因果链：轨迹提炼减少噪声，分层写回提高复用；但论文没有测试错误经验污染与回滚。

知识怎样落到最终模型？

两个真实任务说明 Agent 的价值不是发明新算法，而是把成熟工程技巧组合成完整、稳定的方案。



Fig. 3 | ABLUDA-2 ranks among top-tier human experts in a Kaggle heart disease prediction competition. **a**, First solution in detail. The competition (Kaggle Programming Series 1022) requires predicting the presence of heart disease from 13 clinical features (e.g., age, blood pressure, cholesterol, maximum heart rate, and ST depression) over 100,000 training and 270,000 test rows. ABLUDA-2 produces a heterogeneous gradient boosted decision tree ensemble built from four base models, each trained on the raw clinical features with cross-validation. **a**, CatBoost (15) trained with 10-fold cross-validation, a nearest CatBoost trained with 5-fold cross-validation, an XGBoost (14) trained with 10-fold cross-validation, and a LightGBM (21) trained with 10-fold cross-validation. Each base model's predictions are rank-transformed across all test rows and combined by a 10-fold cross-validation ensemble. **b**, Performance metrics. ABLUDA-2 (blue) achieves an AUC of 0.80429, a log loss of 0.20386, and a private AUC of 0.80429. ABLUDA-1 (orange) achieves an AUC of 0.79402, a log loss of 0.20520, and a private AUC of 0.79402. The ML-Ensemble (yellow) achieves an AUC of 0.78429, a log loss of 0.20602, and a private AUC of 0.78429.

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心脏病：异构树融合

两组 CatBoost 加 XGBoost、LightGBM，分别交叉验证；各模型预测先做 rank transform，再用 OOF 数据搜索融合权重。统一尺度后，校准差异不会直接破坏排序型 AUC 融合。

ADMET：双表征互补

图网络分支学习分子结构，LightGBM 分支读取 ECFP4、FCFP4、MACCS 与 2D 描述符；先逐任务融合两分支，再对 9 个 checkpoint 做受约束加权，兼顾深度表征与经典化学特征。



设计理解

组件协同的意义在误差互补：不同模型提供不同归纳偏置，OOF 权重把选择依据留在训练数据内。

70.7% 证明了什么？

主结果证明完整系统竞争力很强；真实竞赛的匹配设置比较进一步表明，它优于两个知识机制较弱的 Agent。

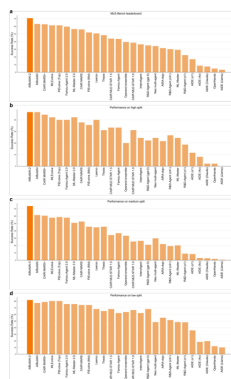


Fig. 2. MLE-Bench results. (a) Overall model size of MLE-Bench. (b) Performance on high-difficulty tasks. (c) Performance on medium-difficulty tasks. (d) Performance on low-difficulty tasks. The y-axis represents the percentage of tasks solved. The x-axis represents the tasks. The bars are orange. The chart shows a general downward trend in performance as task difficulty increases.

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MLE-Bench 主结果

75 项任务总体奖牌率 70.7%；高、中、低难度组分别为 46.67%、73.68%、81.82%，三个分组均排第一。它支持系统跨多模态任务有效，但排行榜基线的底模与预算不完全统一。

同设置基线比较

心脏病私榜 AUC: 0.95529 对 0.95492/0.95425；ADMET MA-RAE: 0.6300 对 0.7000/0.6735。三者使用 Claude Opus 4.7、24 小时和同硬件，整体系统优势更可信。



设计理解

重要边界：这不是 L1/L2 的组件 ablation。架构同时改变，不能把全部增益单独归因于动态加载或经验写回。

离“越用越强”还差什么？

论文证明了一个强系统，却尚未证明知识库会在长期运行中稳定、自主地持续提升。

formatics fingerprints and graph neural network architectures for molecules that are insufficiently covered by generic LLM parametric knowledge or unconstrained web search.

Discussion

Across three evaluation settings spanning general machine-learning benchmarks, a real-world clinical prediction task, and a scientifically demanding drug discovery problem, AIBuildAI-2 attains a level of performance comparable to expert human practitioners: it ranks first on MLE-Bench, places in the top 6.6% of 4,170 human-expert teams on the heart disease prediction competition, and ranks 40th out of 103 teams on the OpenADMET ExpansionK Blind Challenge, all without task-specific customization. Taken together, these results demonstrate that AIBuildAI-2 provides a reliable, generally applicable route for researchers, including those without deep AI engineering expertise, to automatically build competitive AI models for both general benchmark tasks and domain-specific scientific problems.

The strong performance of AIBuildAI-2 can be attributed in large part to its comprehensive, hierarchical, and continually evolving knowledge system, which grounds each design and implementation decision in concrete external expertise rather than the parametric memory of the underlying LLM alone. Building high-performing AI models requires substantial practical engineering know-how, such as which architectures suit a given data modality, which preprocessing stabilizes training, and which failure modes to anticipate during

but only over a narrow, static slice of AI development knowledge. Concurrent work (51) also explores letting an AI development agent accumulate knowledge from its own past runs, but its self-evolution draws solely on the agent's own experience; AIBuildAI-2 instead jointly evolves the knowledge system from both the agent's experience and newly emerging web content, and uses a substantively different hierarchical design. AIBuildAI-2 addresses these limitations simultaneously through breadth (an L2 lookup of around 1,000 curated low-level knowledge documents), hierarchical accessibility (a human-authored L1 index whose values combine high-level knowledge instructions with pointers into L2, retrieved on demand), and self-evolution (post-run takeaways processed by L1 and L2 builders that update both layers), which together drive its consistent gains across MLE-Bench, the heart disease prediction competition, and the OpenADMET drug discovery challenge.

A limitation of AIBuildAI-2 is that, while its knowledge system evolves from both completed runs and newly published web content, the surrounding harness, that is, the prompts and code that define each sub-agent's behavior and their orchestration, remains fixed. Recent agentic systems have begun to explore evolving the harness itself (52, 53), suggesting a natural extension in which AIBuildAI-2 learns from its own runs not only what knowledge to retain but also how to prompt and structure its sub-agents. However, harness-level evolution is intrinsically more difficult than knowledge-level evolution: small changes to prompts or control flow can interact unacceptably with the LLM and

论文原图 · 第 8 页

证据缺口与成本

没有关闭 L1、L2、动态加载、写回的消融，也没有任务数增长曲线、随机种子方差、知识冲突或过时条目测试。复现需要 Opus 4.7、7 条路线、24 小时、24 vCPU、256 GB RAM 和单张 A100。

三个可迁移原则

第一，长期记忆应保存结构化技能而非原始对话；第二，目录常驻、正文按需加载；第三，写回必须带来源、版本、置信度、冲突检测与回滚。只有这样，经验积累才可审计。



设计理解

作者也承认 harness 仍固定：系统学习了“记住什么”，还没有学习“怎样重构自己的工作方式”。

